

ASRA HUSSAIN

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EXPERIENCE

Freelance, Remote: MLOPs Engineer

May 2024 – May 2025

- Optimized model performance and reduced inference latency through quantization, pruning techniques, and multi-threaded serving optimizations.
- Delivered end-to-end MLOps solutions with comprehensive documentation, stakeholder reporting, and knowledge transfer sessions to empower cross-functional teams.

SmartBridge, Hyderabad: Machine Learning Engineer

Aug 2023 – May 2024

- Designed and deployed scalable ML pipelines using Python, scikit-learn, TensorFlow, and Keras with Docker, Kubernetes, and CI/CD. Developed RESTful APIs with FastAPI, Django, and Flask for efficient model serving.
- Enhanced model inference speed by **30%** and reduced response latency by **20%** through advanced model quantization, pruning, and multi-threaded optimizations.

Datagrape.ai, Delhi: Data Analysis Intern

Mar 2023 – Aug 2023

- Built interactive Power BI dashboards using advanced DAX and automated ETL via Power Query.

CERTIFICATIONS

AWS Certified Solutions Architect

2025

Amazon Web Services

EDUCATION

B.Sc

Graduated May 2022

Shri Shivaji Science College

8.1 GPA

PROJECTS

FinQuery RAG Financial Insights Application

Fall 2024

Created a Streamlit-based application that analyzes financial reports and live market data using AI-driven techniques.

- Features a PDF Upload Mode (hybrid retrieval with FAISS, BM25, and transformer-based text splitting) and a Live Data Mode.
- **Technologies:** Python, Streamlit, ChatGrok LLM, Transformer Models, SentenceTransformers, PyTorch, FAISS, BM25, NLP, Alpha Vantage API, LangChain, GenAI, Prompt Engineering,

Agentic AI Neural Logic Engine

Fall 2024

Developed a Python-based research assistant using advanced semantic search algorithms

- Integrated LangChain with OpenAI's conversational AI, DuckDuckGo Search API, and Wikipedia API. Boosted semantic search accuracy by **25%** and reduced query response time by **15%** through iterative tuning.
- **Technologies:** Python, LangChain, GenAI, Prompt Engineering, BM25, Transformer embeddings.
- Click here for the GitHub link

ScrapeGraphAI Web Insights Tool

Fall 2024

Built a Gradio app that scrapes web content and summarizes it using Hugging Face AI models.

- Leverages a smart scraping graph with AI-powered LLM and embedding models to extract concise summaries.
- **Technologies:** Python, Gradio, HuggingFaceEndpoint, HuggingFaceInferenceAPIEmbeddings, SmartScraper-Graph, Playwright, Advanced NLP.

TECHNICAL SKILLS

Programming: Python; Java; R; C++; MySQL; PostgreSQL; FastAPI; Django; Flask; Git

Machine Learning: Python; Scikit-learn; TensorFlow; PyTorch

MLOps: Docker; NLP; LangChain; HuggingFace; LLMs; Transformers; GenAI; Prompt Engineering

Data Analysis: Power BI; Tableau